

Lab Report

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Team: E1

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Goal

1. The goal of this experiment was to test and improve the performance of a line-following vehicle after code modifications made during the previous week.
2. The experiment focused on identifying hardware and software issues that prevented the vehicle from operating correctly and implementing solutions to achieve stable line-following behavior.

Discussion

During the experiment, two major problems were observed:

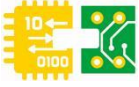
1. The vehicle failed to follow the black line.
2. When the two wheels were lifted off the ground, placing the IR sensors over the black line caused the program to enter the correct functions, and both motors rotated normally. However, when the wheels touched the ground, the vehicle overshot the line while turning and then continued moving straight instead of returning to the line.

Several methods were used to identify and solve the problems:

1. Software Debugging

The code was modified several times to improve the vehicle's performance. The following adjustments were made:

- a. Reduced and adjusted the turning speed.



- b. Corrected the turning direction.
- c. Improved the line-following logic.
- d. Added a condition to stop the vehicle when both IR sensors detected the black line.

2. Motor Inspection

The motors were replaced to ensure that the problem was not caused by faulty motors. After replacement, the motors operated normally, indicating that the motors were not the source of the issue.

3. Speed Optimization

Different motor speeds were tested repeatedly. The vehicle's turning performance improved after multiple adjustments, although it did not completely solve the problem.

4. Wiring Inspection

After checking the hardware connections, mistakes were found in the motor controller wiring. In particular, the connections involving the VIN, 12V power supply, and the breadboard were incorrect.

5. Rewiring

The entire circuit was rewired carefully to ensure that all power and signal connections were correct.

6. Motor controller inspection

The motor controller was replaced to ensure that the problem was not caused by faulty motor controller. However, after replacement, the vehicle can detect and follow a black line on the ground.

Conclusion

Through step-by-step debugging, including code optimization, hardware inspection, speed tuning, and rewiring, our vehicle finally can follow a black line on June 26th.